NTNU

Department of computer science

Examination paper for TDT4175 Information Systems

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Examination time (from-to): 9:00-12:00

Permitted examination support material: *Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book* "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been used. Clean copies of the chapters 1-6 used in the course or of the summary text put out on Black Board are also allowed. *In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.*

Language: English

Number of pages (front page excluded): 3

Number of pages enclosed: 4

Informasjon om trykking av eksamensoppgave	
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Checked by:

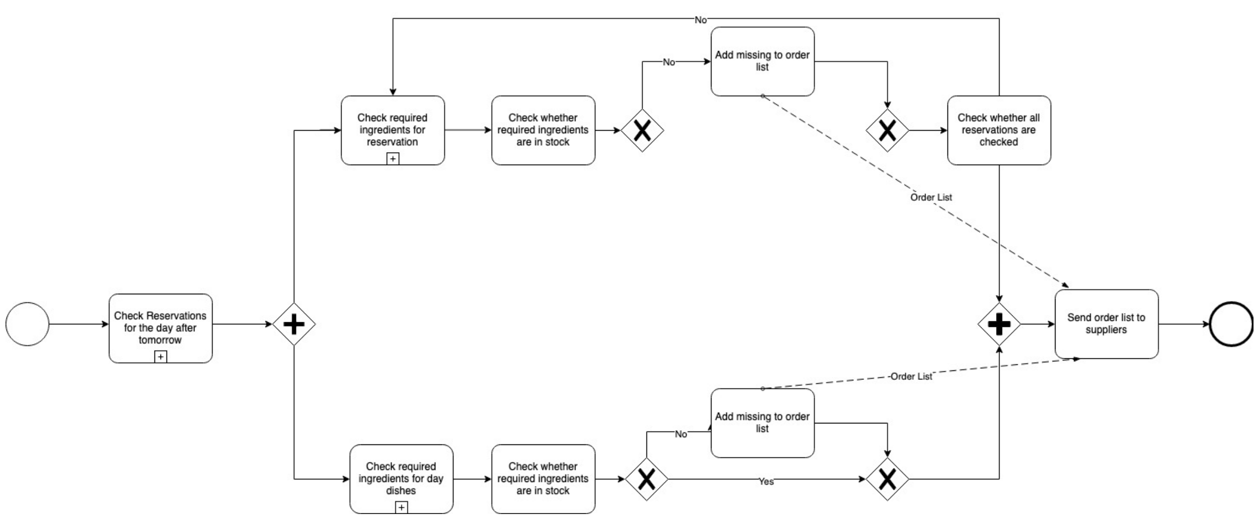
Nana Kwame Henebeng Amagyei

Date Signature

Task 1 - (25 points)

Julie and Eva have opened a new restaurant with the name 'Dine for Life', which is open every day. They aim to serve climate friendly food (vegetarian and bought from local suppliers) and to minimize food waste. To achieve this last objective, they want to buy in just enough ingredients to cover their needs. Customers can come to the restaurant without reservation, but then they have only one choice of food: the dish of the day. They can also make a reservation, at least 2 working days before they want to have dinner. Along with the reservation they must choose what to eat from the menu. When a reservation is made, it is checked whether there is an empty table for the required number of people. If so, the table is reserved. Customers must confirm their visit a day before the reservation date. If no confirmation is received, the reservation is cancelled.
At the end of every working day, the list with reservations for two days later is checked. Based on the required ingredients for the ordered food per reservation, the list of required ingredients for the reservation day is updated. All ingredients that are not in stock are ordered from the local suppliers. The deal is that orders are delivered the next day unless that is a Sunday. In that case the orders are delivered on Monday. Tables not reserved at a given day are open for drop-inn customers. There must be enough ingredients to serve all the not reserved places the dish of the day. Ingredients needed for this are added to the required ingredients list.

Given this case description a student has made the following model.



1.1 (5 points)
The first step of modelling a business process, as advocated by Silver, is choosing the scope of your model. One of the questions to be answered in scoping is: **What does each instance of the process represent?** Answer this question for the model made by the student.

1.2 (20 points)
The model contains several mistakes, both syntactical and semantical. Find at least three syntactical and two semantical mistakes (so, five mistakes in total). Motivate why you consider these to be mistakes.

Task 2 (5 points)

What is a computer-based information system (CBIS)?

Task 3 (5 points)

Describe the difference between data, information and knowledge using examples from the restaurant case.

Task 4 (15 points)

Eva and Julie notice that compiling the order list at the end of every working day is a lot of work when done manually. They work late hours and often the list contains errors because they make mistakes due to being tired. A friend suggests automating (part of) this activity by using an information system. Which type of information system would be suitable to use here? Describe the functionality of the system required to support the activity.

Task 5 (15 points)

After a year Eva and Julie evaluate their achievements with respect to their main business goals. Especially with respect to reducing food waste, they are disappointed. They have the feeling that they had to throw a good deal of ingredients that were unused as well as prepared food but are unsure about the reason for this. They have used a transaction management system that registers reservations, cancelations, payments, orders of ingredients, deliveries of ingredients, payments to personnel, etc.

5.1 (5 points)
What basic transaction processing activities are performed by all transaction processing systems?

5.2 (5 points)
Which kind of information system could be used to get insight in the possible reasons for food waste, using available transaction information?

5.3 (5 points)
Which views (such as the number of no shows per day of the week) could be useful to find out the reason for having to throw food?

Task 6 (15 points)

The order lists are sent till local suppliers by e-post. This has led to delays and misunderstandings (wrong deliveries, for example). Because of these uncertainties the two days ordering window was chosen. For some goods (fresh goods that don't last that long) deliverance at the day of use would be better.

6.1 (7 points)
Which kind of information system could be used to improve the order/delivery process?

6.2 (8 points)
Propose a good way to reorganize the ordering/delivery process using the type of system mentioned in 6.1.